

Assignment 1: Derive the position of maxima and minima for YDSE

Q1. [CO1]

- (a) What are the necessary conditions for good interference pattern?
- (b) Mention some methods to generate coherent sources of light
- (c) Give some examples of interference observed in daily life
- (d) What is the effect on the interference fringes when the screen is moved away from the source in Young's double slit experiment.

Q2. [CO2]

- (a) Why two independent sources of light can never be coherent?
- (b) What would happen if the monochromatic source of light is replaced with white light in Young's double slit experiment.
- (c) How one can locate the central fringe in Fresnel's biprism experiment?
- (d) How would the interference pattern change if the biprism were replaced with a prism having a larger refracting angle?
- (e) If the experiment is performed in a medium other than air (e.g., water), how would the interference pattern be affected?

Q3. [CO3]

- (a) The ratio of intensities of two waves that produce interference is 16:1. Deduce the ratio of maxima to minima intensities for the system. **ANS: 25:9**
- (b) Distance between two slits is 0.1 mm and width of fringes formed on the screen is 5 mm. If the distance between slit and screen is 1 meter, then find the wavelength of light used? **ANS: 5000 Å**
- (c) A biprism of angle 1° and refractive index 1.5 is at a distance of 40 cm from the slit. Find the fringe width at 60 cm from the biprism for sodium light of wavelength 5890 Å. **ANS: 0.08 mm**
- (d) In a biprism experiment using light of wavelength 5890 Å, 40 fringes are observed in the field of view. If the light source is replaced with light of wavelength 4358 Å, then how many fringes will be observed in the field of view. **ANS: 54**
- (e) In a Fresnel biprism experiment, light with a wavelength of 550 nm is used. The distance between the virtual sources is 0.7 mm, and the distance from the biprism to the screen is 1.8 meters. Calculate the position of the 4th order bright fringe. **ANS: 5.657mm from the central maximum**

Extra Practice questions

Q1. A light source emitting lights of two wavelengths namely 400 nm and 500 nm is used in double slit experiment. If the distance between the slits is 0.25 mm and screen is placed at a distance of 1 meter from the slits, then calculate the separation between 2nd order bright fringes due to these two wavelengths. [CO3] **ANS: 0.8mm**

Q2. In a Fresnel biprism experiment, light with a wavelength of 600 nm is used. The distance from the biprism to the screen is 1.5 meters, and the observed fringe width is 0.9 mm. Calculate the distance between the virtual sources. [CO3] **ANS: 1mm**

Q3. In a double slit experiment, light of wavelength 600 nm is used. The distance between the slits is 1 mm, and the screen is placed 2 meters away from the slits. If the distance between slits is changed to 2 mm for 400 nm light, calculate the change in fringe width and determine the new fringe width. [CO3] **ANS: 0.8 mm**

Q4. In a double slit experiment with light of wavelength 480 nm, the distance between the slits is 0.4 mm and the screen is 2 meters away. Calculate the number of fringes that fit into a 10 cm wide region on the screen. **ANS: Approximately 42 fringes**

Q5. In a Young's double slit experiment, the screen is 1 meter away. If the fringe separation is 2 mm for light of wavelength 400 nm, calculate the fringe separation for light of wavelength 600 nm. **ANS: 3 mm**

Solutions of Tutorial no 1

- Ans 1 (a) 1. Light sources should be coherent
2. Monochromatic light produces clear and stable interference fringes.
3. Slits should be narrow
3. Interfering waves should have approx same intensity to produce fringes of good contrast.

(b) YDSE, Lloyd's Mirror, Fresnel biprism

(c) Soap bubbles, oil slicks, CD & DVD, anti reflection coating, wings of butterflies.

(d) ① The fringe width will increase as

$$\beta = \frac{\lambda D}{d}$$

② The overall intensity of fringes may decrease as light spreads out over a larger area on the screen.

③ Angular separation between fringes remains same.

Ans 2 (a) Two independent sources of light can never be coherent because they do not maintain a constant phase relationship with each other over time.

(b) We would get coloured fringes corresponding to different wavelengths present in white light.

(c) Instead of monochromatic source, one can use white light.

(d) $\beta = \frac{D}{d} \lambda = \frac{D \lambda}{2(u-1) \alpha Z_1}$ where α is acute angle of biprism.

If α increases then β decreases.

$$(e) \quad \beta = \frac{\lambda D}{d}, \quad \beta \propto \lambda$$

$$\text{Now } \lambda_w < \lambda_a$$

$$\Rightarrow \beta_w < \beta_a$$

$\Rightarrow$ Fringe width will decrease.

$$\text{Q 3 (a)} \quad I_1 : I_2 = 16 : 1$$

$$\Rightarrow \frac{a_1^2}{a_2^2} = \frac{16}{1} \quad \Rightarrow \frac{a_1}{a_2} = \frac{4}{1}$$

$$\frac{I_{\max}}{I_{\min}} = \frac{(a_1 + a_2)^2}{(a_1 - a_2)^2} = \frac{(4a_2 + a_2)^2}{(4a_2 - a_2)^2} = \frac{25}{9}$$

$$(b) \quad \beta = 5 \text{ mm} = 5 \times 10^{-3} \text{ m}$$

$$d = 0.1 \text{ mm} = 1 \times 10^{-4} \text{ m}$$

$$D = 1 \text{ m}$$

$$\lambda = \frac{\beta d}{D} = \frac{5 \times 10^{-3} \times 1 \times 10^{-4}}{1}$$

$$= 5000 \text{ \AA}$$

$$(c) \quad a = 40 \text{ cm}, \quad b = 60 \text{ cm}$$

$$u = 1.5$$

$$\lambda = 5890 \text{ \AA}$$

$$D = a + b = 1 \text{ m}$$

$$\beta = \frac{\lambda D}{d} = \frac{\lambda D}{2(u-1)a}$$

$$\alpha = 1^\circ = \frac{\pi}{180} \text{ rad}$$

Putting these we get, $\beta = 0.08 \text{ mm}$

(d)

$$\beta = \frac{\lambda D}{d}$$

D & d remain same

$$\lambda_1 = 5890 \text{ \AA}, \quad \lambda_2 = 4358 \text{ \AA}$$

$$N_1 = 40, \quad N_2 = ?$$

Total width of field of view remains the same.

$$\Rightarrow W = N_1 \times \beta_1 = N_2 \times \beta_2$$

$$\Rightarrow N_1 \lambda_1 = N_2 \lambda_2$$

Putting the values, we get $N_2 \approx 54$

(e)

$$y_m = \frac{m \lambda D}{d}$$

$$\lambda = 550 \text{ nm} = 550 \times 10^{-9} \text{ m}$$

$$d = 0.7 \text{ mm} = 7 \times 10^{-4} \text{ m}$$

$$D = 1.8 \text{ m}$$

$$m = 4$$

$$y_4 = \frac{4 \times 550 \times 10^{-9} \times 1.8}{7 \times 10^{-4}}$$

$$\approx 5.66 \text{ mm}$$

Extra Practice Questions

Ans 1

$$y_m = \frac{m \lambda D}{d}$$

$$\lambda_1 = 400 \text{ nm} \quad , \quad d = 0.25 \text{ mm}$$

$$\lambda_2 = 500 \text{ nm} \quad D = 1 \text{ m}$$

$$y_2 \text{ for } \lambda_1 = 3.2 \text{ mm}$$

$$y_2 \text{ for } \lambda_2 = 4 \text{ mm}$$

$$\Delta y \approx 0.8 \text{ mm}$$

Ans 2

$$\beta = \frac{\lambda D}{d}$$

$$\lambda = 600 \text{ nm}$$

$$D = 1.5 \text{ m}$$

$$d = 0.9 \text{ mm}$$

$$\beta = 1 \text{ mm}$$

Ans 3.

$$\beta = \frac{\lambda D}{d}$$

$$\lambda_1 = 600 \text{ nm}$$

$$d_1 = 1 \text{ mm}$$

$$D = 2 \text{ m}$$

$$\lambda_2 = 400 \text{ nm} \quad , \quad d_2 = 2 \text{ mm} \quad , \quad D = 2 \text{ m}$$

$$\beta_1 = 1.2 \text{ mm}$$

$$\beta_2 = 0.4 \text{ mm}$$

$$\begin{aligned} \text{Change in fringe width} &= 1.2 - 0.4 \\ &= 0.8 \text{ mm} \end{aligned}$$

Ans 4.

$$\lambda = 480 \text{ nm} \quad , \quad d = 0.4 \text{ mm} \quad , \quad D = 2 \text{ m}$$

$$\beta = 2.4 \text{ mm}$$

$$\text{Number of fringes} = \frac{\text{Width of region}}{\text{Fringe width}}$$

$$= \frac{100 \text{ mm}}{2.4 \text{ mm}}$$

$$\approx 42$$

Ans 5

$$\beta = \frac{\lambda D}{d}$$

$$\lambda_1 = 400 \text{ nm} , \beta_1 = 2 \text{ mm}$$

$$D = 1 \text{ m} , d = ~~400~~ 0.2 \text{ mm}$$

$$\text{Now } \lambda_2 = 600 \text{ nm}$$

$$\text{For this } \beta = 3 \text{ mm}$$

B. Tech. First Semester (2024-25)

Tutorial Sheet – II

Course: Physics – I (15B11PH111)

Assignment 2: Derivation of wedge-shaped films

Section A: Mandatory Questions

- 1.a) What is the value of phase change upon reflection from medium 1 (refractive index μ_1) to medium 2 (refractive index μ_2) if (i) $\mu_1 > \mu_2$ and (ii) $\mu_1 < \mu_2$?
b) What are the fringes of equal width and fringes of equal inclination?
c) Write the expression for fringe width in interference from thin films. [CO 1]
- 2.a) Using Stoke's relations, show that the excessively thin film appears perfectly dark, when seen by reflected light.
b) Why is the central fringe of Newton's ring formed by reflected light dark?
c) Why are complementary colours visible in interference patterns in thin film for reflected light and transmitted light? [CO 2]
3. Light of wavelength $6000 \text{ \AA}$ falls normally on a thin wedge-shaped film of refractive index 1.4, forming fringes that are 2 mm apart. Find the angle of the wedge. [CO 2]
4. In the Newton's rings arrangement, the radius of curvature of the curved surface is 50 cm. The radii of the 9th and 16th dark rings are 0.18 and 0.2235 cm, respectively. Calculate the wavelength. [CO 2]
5. A soap film of refractive index $\mu = 1.34$ is surrounded by air. Under normal illumination, if a region of the film strongly reflects the wavelength $\lambda = 804 \text{ nm}$, what is the minimum thickness of the film? [CO 3]
6. An equiconvex lens is placed on another equiconvex lens. The radii of curvature of the two surfaces of the upper lens are 50 cm and those of the lower lens are 100 cm. The light waves reflected from the upper and lower surface of the air film (formed between the two lenses) interfere to produce a fringe pattern. If the monochromatic light of wavelength $6000 \text{ \AA}$ is used in the experiment, calculate the radii of the 4th and 16th dark rings. [CO 3]
7. Suppose that the light source in the Newton's ring arrangement emits two wavelengths λ_1 and λ_2 , such that the m -th dark ring due to λ_1 coincides with the $(m + 1)$ -th dark ring due to λ_2 . If R is the radius of curvature of the lens, show that the radius ρ_m of the m -th dark ring due to λ_1 is given by

$$\rho_m = \sqrt{\frac{\lambda_1 \lambda_2 R}{\lambda_1 - \lambda_2}}$$

[CO 4]

8. Consider the formation of Newton's ring with a bi-chromatic light source emitting two closely spaced wavelengths λ_1 and λ_2 . Show that if plano-convex lens is raised by a distance $t_0 = \frac{\lambda_1 \lambda_2}{4(\lambda_1 - \lambda_2)}$, the fringe pattern would disappear. If we continue to move the lens upward, show that the fringe pattern would reappear when lens is raised by a distance $2t_0$. Assume that the light source is placed normally above the lens.

[CO 4]

Section B: Practice Questions

1. In the Newton's ring arrangement, the radius of curvature of the curved surface of the plano-convex lens is 100 cm. For $\lambda = 6 \times 10^{-5}$ cm, what will be the radius of the 9th bright ring?
[CO 2]
2. White light is reflected normally from a soap film of uniform thickness in air. An interference maximum is observed at $\lambda_1 = 0.6 \mu\text{m}$ and a minimum at $\lambda_2 = 0.45 \mu\text{m}$ with no minimum or maximum in between. Assuming $\mu = 1.33$ for the film, calculate the thickness of the film.
[CO 3]
3. A microscope of refractive index $\mu = 1.55$ is to be coated with a MgF_2 film ($\mu = 1.38$) to increase transmission of normally incident light $\lambda = 550$ nm. What should be the minimum thickness of the film?
[CO 4]
4. In the Newton's rings arrangement, if the incident light consists of two wavelengths of 4000 and 4002 Å, calculate the distance (from the point of contact) at which the rings will disappear. Assume that the radius of curvature of the lens is 400 cm.
[CO 4]

1. see slides/text book
2. see slides/text book

3. $\beta = 2 \text{ mm} = 2 \times 10^{-3} \text{ m}$
 $\lambda = 6000 \text{ \AA} = 6 \times 10^{-7} \text{ m}$
 $\mu = 1.4$

for interference in wedge shaped film $\rightarrow$

$$\beta \approx \frac{\lambda}{2\mu\theta}$$

Where $\theta \rightarrow$ angle of the wedge.

$$\Rightarrow \theta \approx \frac{\lambda}{2\mu\beta} = 1.071 \times 10^{-4} \text{ radian}$$

4. $\lambda = \frac{\tilde{\nu}_{16} - \tilde{\nu}_9}{4 \times 7 R} = \frac{4(\tilde{\nu}_{16} - \tilde{\nu}_9)}{4 \times 7 R} = \frac{\tilde{\nu}_{16} - \tilde{\nu}_9}{7 R}$
 $= \frac{(0.2235) - (0.18)}{350} \text{ cm} = 5015 \text{ \AA}$

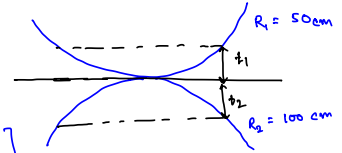
for max reflection,
 phase diff due to path diff b/w $= \pi$

two reflected waves from two
 surfaces of the soap film

$$\Rightarrow \frac{2\pi}{\lambda} \mu \cdot 2t = \pi \quad ; \quad t \rightarrow \text{thickness of the film}$$

$$\Rightarrow t = \frac{\lambda}{4\mu} = 150 \text{ nm}$$

6. $2t_1 R_1 = r_m^{\sim}$
 $2t_2 R_2 = r_m^{\sim}$



$$\Rightarrow 2(t_1 + t_2) = r_m^{\sim} \left[\frac{1}{R_1} + \frac{1}{R_2} \right]$$

$= m\lambda$ [for dark ring]

$$\Rightarrow r_m = \sqrt{\frac{R_1 R_2}{R_1 + R_2} m\lambda} = 0.0447 \sqrt{m} \text{ cm}$$

$$\Rightarrow r_4 = 0.0894 \text{ cm} \quad \& \quad r_{16} = 0.1788 \text{ cm}$$

7. $2Rt = r_m^{\sim}$ $\left\{ \begin{array}{l} 2t = m\lambda_1 = (m+1)\lambda_2 \\ \Rightarrow m(\lambda_1 - \lambda_2) = \lambda_2 \\ \Rightarrow m = \frac{\lambda_2}{\lambda_1 - \lambda_2} \end{array} \right.$
 $\Rightarrow \frac{r_m^{\sim}}{R} = 2t = m\lambda_1$
 $= \frac{\lambda_1 \lambda_2}{\lambda_1 - \lambda_2}$
 $\Rightarrow r_m^{\sim} = \frac{\lambda_1 \lambda_2 R}{\lambda_1 - \lambda_2} \Rightarrow r_m = \sqrt{\frac{\lambda_1 \lambda_2 R}{\lambda_1 - \lambda_2}}$

8. The height at which fringes disappear will be $\frac{\lambda_1 \lambda_2}{4(\lambda_1 - \lambda_2)}$
 $\frac{2t_0}{\lambda_2} - \frac{2t_0}{\lambda_1} = \frac{1}{2} \Rightarrow 2t_0 \frac{\lambda_1 - \lambda_2}{\lambda_1 \lambda_2} = \frac{1}{2} \Rightarrow t_0 = \frac{\lambda_1 \lambda_2}{4(\lambda_1 - \lambda_2)}$
 for fringes to reappear: $\frac{2t}{\lambda_2} - \frac{2t}{\lambda_1} = 1 \Rightarrow t = \frac{\lambda_1 \lambda_2}{2(\lambda_1 - \lambda_2)} = 2t_0$

Section B: Practice Question

4. $\lambda_2 = 4600 \text{ \AA}$ $\lambda_1 = 4002 \text{ \AA}$

$$t_0 = \frac{\lambda_1 \lambda_2}{4(\lambda_1 - \lambda_2)} \approx 0.02 \text{ cm}$$

1. $2Rt = r_m^{\sim}$
 $\Rightarrow 2t = \frac{r_m^{\sim}}{R} = (m + \frac{1}{2}) \lambda$
 $\Rightarrow r_m = \sqrt{(m + \frac{1}{2}) \lambda R}$
 $= \sqrt{\frac{(2m+1) \lambda R}{2}} = \sqrt{\frac{19 \times 100 \times 10^{-5}}{2}}$
 $= \sqrt{57 \times 10^{-3}}$

for maximum,

$$\frac{2\pi}{\lambda_1} 2\mu t = (2m+1)\pi \rightarrow (0)$$

& for minimum,

$$\frac{2\pi}{\lambda_2} 2\mu t = (2m+2)\pi \rightarrow (2)$$

$$\Rightarrow 4\pi \mu t \left(\frac{1}{\lambda_2} - \frac{1}{\lambda_1} \right) = \pi \quad [b_1 (2) - (0)]$$

$$\Rightarrow t = \frac{\lambda_1 \lambda_2}{\lambda_1 - \lambda_2} \cdot \frac{1}{4\mu} \approx 338.34 \text{ nm}$$

3. thickness $= \frac{\lambda_0}{4\mu} = \frac{550}{4 \times 1.38} \text{ nm} = 99.6 \text{ nm}$

Tutorial 3 Physics-1 (15B11PH111)
ODD2024 (2024-2025)

Q1. [CO1]

- (a) What is the purpose of a half-silvered mirror in Michelson Interferometer?
- (b) Write down the conditions of getting circular, straight and curved fringes in Michelson Interferometer.
- (c) How will you differentiate the circular fringe pattern formed in Newton's ring and Michelson Interferometer?
- (d) Write some applications of Michelson Interferometer.
- (e) What is the difference between Fraunhofer and Fresnel diffraction?
- (f) Which wavefront of light is more suitable for observing single slit diffraction pattern?

Q2. [CO2]

- (a) Show that in Michelson Interferometer fringes collapse at centre as the spacing between the mirror M1 and image of mirror M2 (i.e. d) decreases.
- (b) Show that in single slit diffraction pattern, the intensity of third order secondary maxima (I_3) is 0.83 % of principal maxima intensity (I_0).
- (c) A single slit is illuminated by light composed of two wavelengths λ_1 and λ_2 . One observes that due to Fraunhofer diffraction the first minima obtained for λ_1 coincides with the third diffraction minima of λ_2 . What is the relation between λ_1 and λ_2 ?

Q3. [CO3]

- (a) A light having two closely spaced wavelengths of 5000Å and 5005Å is incident of Michelson interferometer. Through what distance a mirror should be moved between two successive appearances of fringes. [Ans: 2.5×10^{-4} m]
- (b) In one arm of a Michelson interferometer, a glass chamber is placed with attachments for evacuating the inside and putting gases in it. The space inside the container is 2 cm wide. Initially, the container is empty. As gas is slowly let into the chamber, you observe that dark fringes move past a reference line in the field of observation. By the time the chamber is filled to the desired pressure, you have counted 122 fringes move past the reference line. The wavelength of the light used is 632.8 nm. What is the refractive index of this gas? [ANS: $\mu = 1.0019$].
- (c) Visible light of wavelength 550 nm falls on a single slit and produces its second diffraction minimum at an angle of 45.0° relative to the incident direction of the light. Determine the slit width and the angle at which first minimum occurs. [ANS: $d = 1.56 \mu\text{m}$; 20.7°].
- (d) Suppose that in Young's experiment, slits of width 0.020 mm are separated by 0.20 mm. If the slits are illuminated by monochromatic light of wavelength 500 nm, how many bright fringes are observed in the central peak of the diffraction pattern? [ANS: 19].
- (e) A single slit of width $10\mu\text{m}$ is illuminated by light of wavelength 100 nm. What is the width of central maxima and second maximum, if the distance of screen from slit is 50 cm? [ANS: 10mm, 5mm].
- (f) In a double slit setup, the width of the transparent part is 0.02 cm and the opaque part is 0.04cm. The two slits are illuminated by a monochromatic source of $\lambda = 6 \times 10^{-5}$ cm. The distance of slits to the screen is 10 cm. Then (i) calculate the positions of maxima inside the first diffraction minima and (ii) find the missing orders of spectra. [ANS: (i) 0.01cm, 0.02cm; (ii) 3, 6, 9, 12..... will be missing order].

Tutorial - 3 Solution
Physics - 1 (ISBII PHIII) ODD 2024

Q1 [C01]

(a). To divide a light beam into two equal parts.

(b) # If mirror M_1 & M_2 are perpendicular ($\perp$) $\rightarrow$ Circular fringes.

If mirror M_1 & M_2 are not exactly perpendicular i.e. one of the mirror is tilted and distance d between the mirrors is small $\rightarrow$ localized fringes (straight)

If $M_1 \nparallel M_2$ and d has appreciable value $\rightarrow$ localized (Curved fringes)

(c) (i) In Newton's ring, the fringes is formed due to variable thickness and constant inclination while in Michelson Interferometer, the constant thickness and variable inclination is responsible for the formation of fringes.

(ii). Newton's ring have minimum order at the centre while Michelson's interferometer have maximum order at the centre.

(d) Applications of Michelson Interferometer are: -

(i) To measure the speed of light in different mediums.

(ii). Precise measurement of distance of less than micron order.

(iii). Precise measurement of refractive index or thickness of transparent medium.

(ii) Difference between two neighboring spectral lines.

(e) (i) In Fresnel diffraction, the source of light and screen is kept at a finite distance while the distance between source and screen is infinite in case of Fraunhofer diffraction.

(ii) In Fresnel diffraction spherical or cylindrical wavefronts are considered while only plane wavefront are used in Fraunhofer diffraction.

(iii) Lenses are not used in Fresnel diffraction while convex lenses are used in Fraunhofer diffraction.

(d) Plane ~~wavefront~~ wavefront.

Q2. [C02]

In Michelson Interferometer

(a) For dark fringe

$$2d \cos \theta = n\lambda$$

$$\theta = \cos^{-1} \left(\frac{n\lambda}{2d} \right)$$

$$n = \frac{2d \cos \theta}{\lambda}$$

if λ fixed
 $d \downarrow \quad n \downarrow$

Let us ~~take~~ consider $\lambda = 500 \text{ nm}$.
 $d = 0.025 \text{ cm}$.

$$\theta = \cos^{-1}(0.001n).$$

$$\theta = 0, 2.56, 3.63, 4.44, 5.13, 5.73$$

$$n = 1000, 999, 998, 997, \dots$$

$$\text{if } d = 0.0025 \text{ cm.}$$

$$\theta = 0, 8.11, 11.48, 14.07$$

$$n = 100, 99, 98, 97$$

$\Rightarrow$ which confirms that the fringes collapse at centre if d decreases.

$$(b) \text{ Intensity } I = A^2 \frac{\sin^2 \beta}{\beta^2}$$

for principal maxima

$$\beta = 0.$$

$$I = I_0 = A^2$$

for third order secondary maxima.

$$\beta = \pm(2n+1)\frac{\pi}{2} = \pm \frac{7\pi}{2}$$

$$I_3 = A^2 \frac{\sin^2 \frac{7\pi}{2}}{(\frac{7\pi}{2})^2} = 0.0083 I_0$$

$$= 0.83\% \text{ of } I_0$$

(C)

Condition for minima.

$$b \sin \theta = \pm n \lambda$$

$$n = 1, 2, 3, \dots$$

first minima for λ_1 wavelength

$$b \sin \theta = \pm \lambda_1$$

third minima for λ_2 wavelength

$$b \sin \theta = \pm 3\lambda_2$$

$$\Rightarrow \boxed{\lambda_1 = 3\lambda_2}$$

Q3 [C03]

(a).

$$\Delta d = \frac{\lambda_1 \lambda_2}{2(\lambda_1 - \lambda_2)}$$

$$= \frac{5000 \times 10^{-10} \times 5005 \times 10^{-10}}{2 \times (5005 - 5000) \times 10^{-10}}$$

$$= 2.5 \times 10^{-4} \text{ meter.}$$

$$(b) \lambda = 632.8 \text{ nm.}$$

$$2(n-1)t = (m-n)\lambda.$$

$$m-n = 122.$$

$$\mu = 1 + \frac{122 \times 632.8 \times 10^{-9}}{2 \times 2 \times 10^{-2}}$$

$$\boxed{\mu = 1.0019}$$

$$(C). \lambda = 550 \text{ nm, } m=2, \theta_2 = 45.0^\circ$$

$$b \sin \theta = m \lambda$$

$$b = \frac{m \lambda}{\sin \theta} = \frac{2 \times 550 \times 10^{-9}}{\sin 45^\circ}$$

$$b = \frac{1100 \times 10^{-9}}{0.707} = 1.56 \times 10^{-6} \text{ m.}$$

$$= 1.56 \mu\text{m.}$$

$$\sin \theta_1 = \frac{550 \times 10^{-9}}{1.56 \times 10^{-6}}$$

$$= \underline{\underline{20.7^\circ}}$$

(d)

$$b = 0.020 \text{ mm.}$$

$$d = 0.2 \text{ mm.}$$

$$\frac{d}{b} = \frac{0.2}{0.02} = 10$$

i.e. 10th order interference maxima will coincide with first order

diffraction minima.

So the number of bright fringes observed in central peak of diffraction pattern

$$= (2n-1) = 2 \times 10^{-1} \\ = \underline{\underline{19}}$$

(e)

$$b = 10^{-5} \text{ m}, \lambda = 100 \text{ nm} = 10^{-7} \text{ m} \\ D = 50 \text{ cm} = 0.5 \text{ m}.$$

for first minima

$$\theta_1 = \sin^{-1} \left(\frac{\lambda}{b} \right) = \sin^{-1} \left(\frac{10^{-7}}{10^{-5}} \right)$$

$$\theta_1 = 0.01 \text{ radian}.$$

Angular width ~~= $2\theta_1$~~ of Central maxima = $2\theta_1 = 2 \times 0.01 = 0.02 \text{ rad}.$
width of central maxima

$$= 2D\theta_1 = 2 \times 0.5 \times 0.01 \\ = 0.01 \text{ meter} = \underline{\underline{1 \text{ cm}}} \\ = \underline{\underline{10 \text{ mm}}}$$

Second maxima will occur between 2nd order ~~maxima~~ minima and 3rd order minima.

Position of Second order minima.

$$\theta_2 = \sin^{-1} \left(\frac{2\lambda}{b} \right) = 0.02 \text{ rad}.$$

Position of 3rd order minima.

$$\theta_3 = \sin^{-1} \left(\frac{3\lambda}{b} \right) = 0.03 \text{ rad}.$$

Angular width of Second order maxima = $\theta_3 - \theta_2 = 0.01 \text{ rad}.$

width of second order maxima

$$= D(\theta_3 - \theta_2) \\ = 0.5 \times 0.01 = 0.005 \text{ m} \\ = \underline{\underline{5 \text{ mm}}}.$$

(f)

$$b = 0.02 \text{ cm}, a = 0.04 \text{ cm} \\ d = a + b = \underline{\underline{0.06 \text{ cm}}} \\ D = 10 \text{ cm}, \lambda = 6 \times 10^{-5} \text{ cm}.$$

first minima.

$$b \sin \theta_1 = \lambda \Rightarrow \sin \theta_1 = \frac{\lambda}{b} = \frac{6 \times 10^{-5}}{0.02 \times 10^{-2}} \\ = 3 \times 10^{-3}.$$

nth order interference maxima.

$$d \sin \theta'_n = n\lambda \Rightarrow \sin \theta'_n = \frac{n\lambda}{d} \\ = \frac{n \times 6 \times 10^{-5}}{0.06 \times 10^{-2}} = 10^{-3} n.$$

$$\theta'_1 = 10^{-3} \text{ rad.}$$

$$\theta'_2 = 2 \times 10^{-3} \text{ rad}$$

$$\theta'_3 = 3 \times 10^{-3} \text{ rad}$$

$$\theta'_4 = 4 \times 10^{-3} \text{ rad}$$

$$\theta'_5 = 5 \times 10^{-3} \text{ rad}$$

$$\theta'_6 = 6 \times 10^{-3} \text{ rad}$$

$$\theta'_7 = 7 \times 10^{-3} \text{ rad}$$

$$\theta'_8 = 8 \times 10^{-3} \text{ rad}$$

$$\theta'_9 = 9 \times 10^{-3} \text{ rad}$$

$$\theta'_{10} = 10 \times 10^{-3} \text{ rad}$$

$$\theta'_{11} = 11 \times 10^{-3} \text{ rad}$$

$$\theta'_{12} = 12 \times 10^{-3} \text{ rad}$$

$$\theta'_{13} = 13 \times 10^{-3} \text{ rad}$$

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$$\theta'_{16$$

Tutorial-4 Physics-1 (15B11PH111)

Session: 2024-25

CO1 (Remembering)

1. Explain the followings:
 - a. Grating element
 - b. Negative and positive crystals
 - c. Brewster's angle
 - d. Brewster's law
 - e. Malus law

2. What is double refraction?

CO2 (Understanding)

3. Why is the grating preferred over single or double slit system to observe the diffraction pattern?
4. What are missing orders in N-slit diffraction? Why do these missing diffraction orders are observed?

CO3 (Analysing)

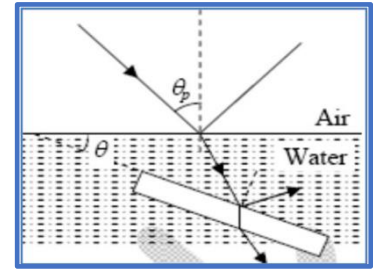
5. Diffraction gratings with 10,000 lines per cm are readily available. Suppose you have one, and you send a beam of white light through it to a screen which is 2 m away from the grating. Find the:
 - a. Angles for the first-order diffraction correspond to the shortest and longest wavelengths of visible light (380 and 760 nm, respectively). **[Ans: $\theta_V = 22.33^\circ$, $\theta_R = 49.46^\circ$]**
 - b. Distance between the ends of the rainbow of visible light obtained on the screen for first order diffraction. **[Ans: 1.52 m]**
6. What should be the minimum number of lines per cm in a 2.5 cm wide grating which will just resolve two emission sodium lines (5890 Å and 5896 Å) in the first order diffraction? **[Ans: 392]**
7. Determine the angle of refraction and polarization angle of the polarizer if the refractive index of the polarizer is 1.73. **[Ans: $\theta_P = 60^\circ$, $\theta_R = 30^\circ$]**
8. An unpolarized beam of light is passed through a set of two plane polarizers, placed back-to-back. What should be the relative orientation of their transmission axes so that the beam exiting from the second polarizer has irradiance of (i) $I_0/2$ and (ii) $I_0/4$? **[Ans: 0° , 45°]**

CO4 (Applying)

9. Draw the intensity pattern for a 4-slit diffraction pattern in which separation between slits is three times the width of the slit.
 10. Two linear polarizers are kept in such a way that their transmission axes are at right angle to each other. What should be the orientation of the transmission axis third polarizer so that the if an unpolarized is passed through first polarizer, the intensity of exiting beam is $I_0/8$? **[Ans: 45°]**
-

Extra Problems

11. Find the number of slits per cm for monochromatic light of the wavelength of 600 nm, which strikes a grating and produces the fourth-order bright line at a 30° angle. [CO3] [Ans: 2080]
12. A plane transmission grating has 40000 lines and $(e+d) = 12.5 \times 10^{-5}$ cm. Calculate the maximum resolving power, if the wavelength of light used is 5000 Å. [CO3] [Ans: 8000]
13. A diffraction grating used at normal incidence gives a yellow line ($\lambda=6000$ Å) in a certain spectral order superimposed on a blue line ($\lambda=4800$ Å) of next higher order. If the angle of diffraction is $\sin^{-1}(3/4)$. Calculate the grating element. [CO3] [Ans: $3.2 \mu\text{m}$]
14. Light incident at polarizing angle θ_p on a water surface ($n=1.33$). It further refracted into the water medium and strikes a submerged glass slab ($n=1.50$) as shown in Figure. If the light reflected from the upper surface of the slab is completely polarized, then determine angle (θ) between the water surface and the glass slab. [CO4] [Ans: 11.5°]



Solution Tutorial-4
Physics-1 (15B11PH111)
Session: 2024-25

1 (a) Grating element

In a transmission grating the separation between two consecutive slits, is known as grating element.

(b) Negative and Positive crystals

For a refracting crystal
if $v_e > v_o$
or $\mu_e < \mu_o$ } Negative crystal
and

if $v_o > v_e$
or $\mu_o < \mu_e$ } Positive crystal

(c) Brewster angle

It is the angle at which, if an unpolarized ray is incident, the reflected ray is completely polarized, perpendicular to the plane of incidence.

(d) Brewster law

According to the Brewster law, the refractive index of medium is equal to the tangent of the polarizing angle.

$$\text{i.e. } \mu = \tan \theta$$

$\theta \rightarrow$ Polarizing angle.

(e) Malus law

Malus law states that the intensity of a plane polarized light, that passes through an analyzer is directly proportional to the square of cosine of the angle between the plane of polarization of light and the transmission axis of the analyzer.

2- Double refraction is an optical phenomenon in which, an unpolarized light when enters in an isotropic medium, it splits into two rays travelling (in general) in different directions.

3- The intensity of the diffracted beam is directly proportional to the number of slits involved in the diffraction. So, by using a diffraction grating, we can get the diffraction pattern of much larger intensity as compared to that ~~observed~~ obtained by single or double slit diffraction.

4- When we observe the diffraction through double slit, the diffraction pattern will be a combination of
 i- single slit diffraction
 ii- double slit interference.
 Resultant diffraction pattern is the resultant of these two phenomena.

Now, condition for the interference maxima is -
 $d \sin \theta = m \lambda$ - (1)

and condition for the diffraction minima -

$$b \sin \theta = n \lambda \quad - (2)$$

~~If θ and λ are~~

If θ , λ , d and b are adjusted in such a way that (1) and (2) both are satisfied, then for certain interference maxima will be missing from the double slit diffraction pattern as these will be ~~sw~~ swallowed up by the diffraction minima.

This is called missing orders in diffraction pattern

$$5(a) d = \frac{1}{10000} \text{ cm} = 10^{-6} \text{ m}$$

Now for $\lambda_v = 300 \text{ nm}$ (Violet)

$$\sin \theta_v = \frac{m \lambda_v}{d}$$

$$\text{or } \sin \theta_v = \frac{1 \times 3.0 \times 10^{-7}}{1 \times 10^{-6}}$$

$$= 0.3$$

$$\theta_v = 22.33^\circ$$

for $\lambda = 760 \text{ nm}$ (red)

$$\sin \theta_R = \frac{m \lambda_R}{d}$$

$$= \frac{1 \times 7.6 \times 10^{-7}}{1 \times 10^{-6}}$$

$$= 0.76$$

$$\theta_R = 49.46^\circ$$

(b)-

Clearly,

$$y_R = x \tan \theta_R$$

$$= 2 \times \tan(49.46^\circ)$$

$$= 2.338 \text{ m}$$

and

$$y_v = x \tan \theta_v$$

$$= 2 \times \tan(22.33^\circ)$$

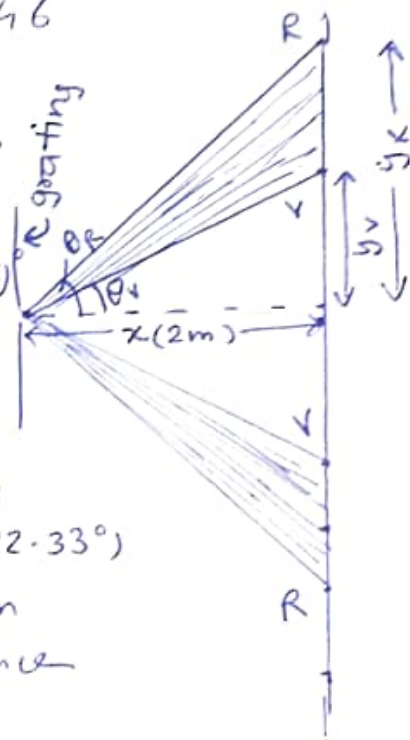
$$= 0.815 \text{ m}$$

Required distance

$$\Delta y = y_R - y_v$$

$$= 2.338 - 0.815$$

$$\Delta y = 1.523 \text{ m}$$



6- We know $\frac{\lambda_m}{d \lambda} = n N$

$$\lambda_{\text{mei}} = \frac{\lambda_1 + \lambda_2}{2} = 5.893 \times 10^{-7} \text{ m}$$

$$d \lambda = 6 \times 10^{-10} \text{ m}$$

$$\text{So, } \frac{5.893 \times 10^{-7}}{6 \times 10^{-10}} = 1 \times N$$

$$N = 981.66$$

No. of slits should not be less than 981.66.
So, practically, minimum no. of slits required = 982

and no. of slits per cm

$$= \frac{982}{2.5} = 392.8$$

7. $\tan \theta_B = 1$

$\tan \theta_B = 1.73$

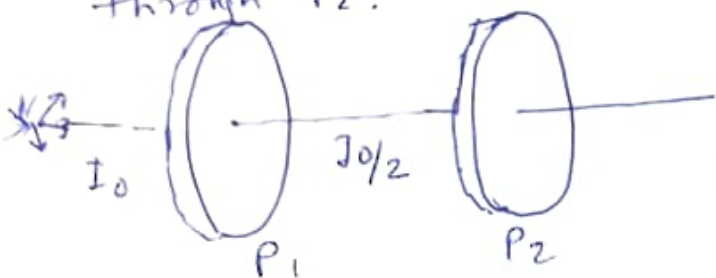
$\theta_B = 60^\circ$

also, $\theta_B + r = 90^\circ$

So, $r = 30^\circ$

8- (a). When the unpolarized light will pass through P_1 , the intensity will reduce to $I_0/2$. The intensity will reduce as

$$I = \frac{I_0}{2} \cos^2 \theta$$
 on passing through P_2 .



(a) for emergence intensity to be $I_0/2$, the formula reduces to.

$$\frac{I_0}{2} = \frac{I_0}{2} \cos^2 \theta$$

$\Rightarrow \cos^2 \theta = 1 \quad \theta = 0^\circ$

b- for emergence intensity (after passing through P_2) to be $I_0/4$, the formula reduces to.

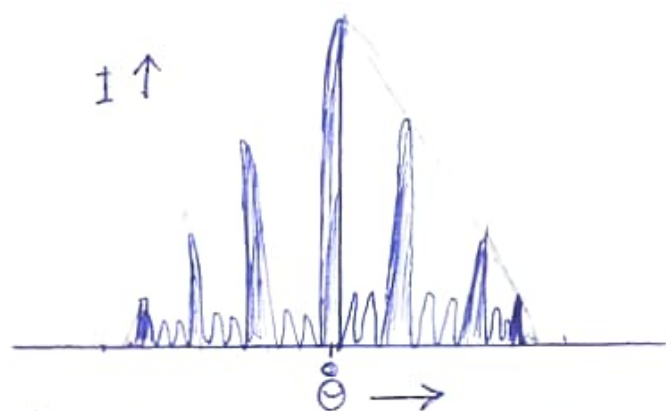
$$\frac{I_0}{4} = \frac{I_0}{2} \cos^2 \theta$$

$$\cos^2 \theta = \frac{1}{2}$$

$$\cos \theta = \frac{1}{\sqrt{2}}$$

$$\theta = 45^\circ$$

9-



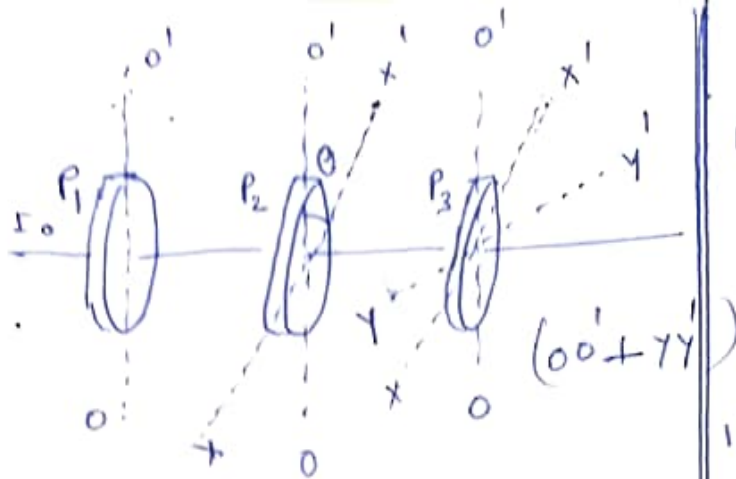
$\frac{d}{b} = \frac{1}{2}$, so: $\left[2 \left(\frac{d}{b} \right) - 1 \right] = 7$

peaks within ~~principle~~ ~~maxima~~ central maxima

• $N = 4$, so $(N - 2) = 4$ maxima between any two consecutive maxima

• $N = 4$, so $(N - 1) = 3$ minima between any two consecutive maxima.

10- Let us assume that polarizer P_2 should be placed in such a way that the transmission axis makes an angle θ with respect to that of P_1 .



- Intensity of light after passing $P_1 = I_0/2$
- Intensity of light after passing $P_2 = \frac{I_0}{2} \cos^2 \theta$
- Intensity of light after passing through P_3

$$= \left(\frac{I_0}{2} \cos^2 \theta \right) \cos^2 (90 - \theta)$$

$$= \frac{I_0}{2} \cos^2 \theta \cdot \sin^2 \theta$$

$$= \frac{I_0}{8} \sin^2 (2\theta)$$

as per question.

$$\frac{I_0}{8} \sin^2 (2\theta) = \frac{I_0}{8}$$

$$\text{so } \sin^2 (2\theta) = 1$$

$$\sin (2\theta) = 1 = \sin 90^\circ$$

$$2\theta = 90^\circ$$

$$\theta = 45^\circ$$

Extra Problems

11- $d \sin \theta = n \lambda$

solving $d = 4.8 \times 10^{-6} \text{ m}$

So No. of slits per cm $= \frac{1}{d}$
 $= 2000$

12- $n_{\max} = \frac{a+b}{\lambda} = 2.5$

Hence max. orders possible = 2

So Max. resolving power
 $= n_{\max} N$

$$= 2 \times 40000$$

$$= 80,000$$

13- Position of n^{th} order spect for yellow.

$$(a+b) \sin \theta = n \lambda \quad (1)$$

Position of $(n+1)^{\text{th}}$ order spectra for blue

$$(a+b) \sin \theta = (n+1) \lambda_B \quad (2)$$

Since they overlap.

Substituting solving (1) and (2)

$$n = 4$$

Substituting $n = 4$ and $\sin \theta = 3/4$ in

eqn. (1) $(a+b) = 3.2 \text{ nm}$

14- for air-water interface

$$\theta_{P_1} = \tan^{-1} \left(\frac{1.33}{1} \right) = 53.1^\circ$$

$$\text{So } \theta_r = 90 - 53.1 = 36.9^\circ$$

for water-glass interface

$$\theta_{P_2} = \tan^{-1} \left(\frac{1.5}{1.33} \right) = 48.4^\circ$$

angle between surface

$$= \theta_{P_2} - \theta_r = 11.5^\circ$$

B. Tech. First Semester (2024-25)

Physics – 1(15B11PH111), Tutorial Sheet – 5

Assignment 3: Briefly discuss the construction and working of bi-quartz polarimeter.

1. Calculate the thickness of a half wave plate of quartz for $\lambda = 6 \times 10^{-7}$ cm? Given $\mu_e = 1.553$, $\mu_o = 1.544$. Also calculate the thickness of quarter wave plate. [CO2]
2. A plate of thickness 0.020 mm is cut from calcite with optic axis parallel to the face. Given $\mu_o = 1.648$ and $\mu_e = 1.481$, find out those wavelengths in the range 400 to 780 nm for which the plate behaves as a HWP and for which the plate behaves as a QWP. [CO2]
3. A QWP has thickness of 0.002 cm. It is cut with faces parallel to optic axis, $\mu_e = 1.592$, $\mu_o = 1.586$. Monochromatic light incident normally on the plate emerges as (i) LCP light, (ii) RCP light. Determine the wavelengths of light in each case assuming that there is no variation of refractive indices with wavelength. [CO3]
4. A transparent plate is given. Using two Nicol prisms how would you find whether the given plate is QWP, HWP or simple glass plate. [CO2]
5. A quartz plate cut perpendicular to the optic axis is to be used to rotate the plane of polarized light through an angle of 90° . If the light to be used is the green light of wavelength 5461 Å, find its thickness. [CO3]
6. If 20 cm length of a certain solution causes right handed rotation of 42° and 30 cm length of another solution causes left handed rotation of 27° . What optical rotation will be caused by 30 cm length of a mixture of the above solutions in the volume ratio 1 : 2. The solutions are not chemically active. [CO4]
7. Plane polarized light passes through a quartz plate with its optic axis parallel to the refracting faces. Calculate the least thickness of the plate for which the emergent beam will be plane polarized, given $\mu_o = 1.5442$, $\mu_e = 1.5533$ for $\lambda = 5893$ Å [CO3]
8. A sample of sugar is mixed with small amount of NaCl. The impure sugar weighing 80 gm is dissolved in a litre of water. The solution gives an optical rotation of 9.9° when polarimeter tube of length 20 cm is used. If the specific rotation of pure sugar solution is $66^\circ \text{ dm}^{-1} (\text{gm/cc})^{-1}$. Find the percentage purity of the sugar sample. [CO3]

Additional Questions [CO4]

1. A QWP is rotated between two crossed polaroids. If an unpolarized beam is incident on the first polaroids, discuss the variation of intensity of the emergent beam as the QWP is rotated. What will happen if we have a half wave instead of a QWP.
2. In figure below, suppose that the transmission axes of the left (P1) and right (P3) polarizing disks are perpendicular to each other, also, let center disk be rotated on the common axis with an angular speed ω . Show that if unpolarized light is incident on the left disk with intensity, the intensity of the beam emerging from right disk is
$$I = \frac{I_{\max}}{16} (1 - \cos 4\omega t)$$
3. Consider a linearly polarized beam propagating in + z direction incident normally on a calcite or quartz crystal whose optic axis is along the direction and the oscillating electric field is assumed to make an angle ϕ with y axis as shown in the figure below, here $\phi = 45^\circ$. Find the value of the thickness (d) for which it will act as a QWP or as a HWP.

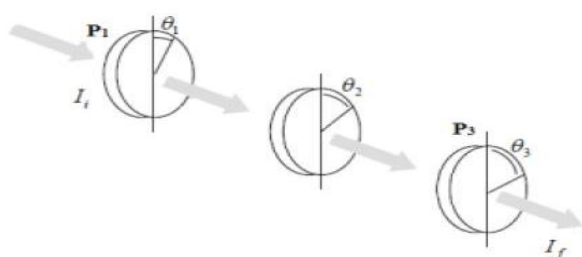


Fig. of Q. 2

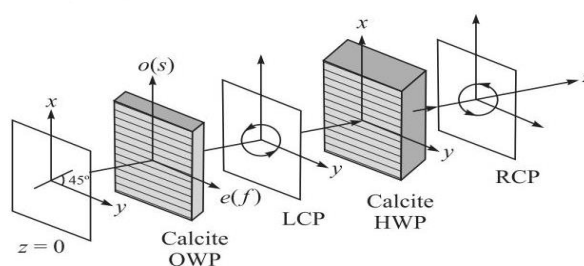


Fig. of Q. 3

B. Tech. First Semester (2024-25)
Physics – 1(15B11PH111), Tutorial Sheet – 5

Solution of Tutorial 5 Physics-I

$$(1) (\mu_e \mu_o) t = \frac{\lambda}{2} \Rightarrow t = \frac{\lambda}{2(\mu_e \mu_o)} = 33.33 \mu m$$

$$(\mu_e \mu_o) t = \frac{\lambda}{4} \Rightarrow t = \frac{\lambda}{4(\mu_e \mu_o)} = 16.66 \mu m$$

$$(2) \text{HWP } \lambda = \frac{2(\mu_o \mu_e) t}{(2m+1)} = \frac{66800 \text{ \AA}}{(2m+1)} \quad m = 4, 5, 6, 7$$

$$\lambda = 7422.22 \text{ \AA}, 6072.72 \text{ \AA}, 5138.46 \text{ \AA}, 4453.33 \text{ \AA}$$

$$\text{QWP } \lambda = \frac{4(\mu_o \mu_e) t}{(2m+1)} = \frac{133600 \text{ \AA}}{(2m+1)}$$

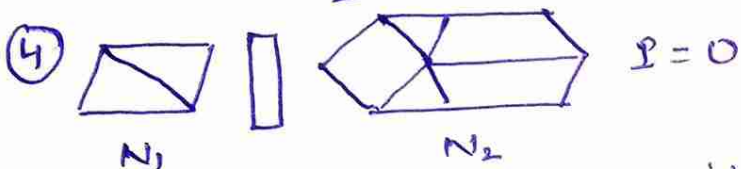
$$m = 9, 10, 11, 12, 13, 14, 15, 16$$

$$\lambda = 7031.57 \text{ \AA}, 6361.90 \text{ \AA}, 5808.69 \text{ \AA}, 5344 \text{ \AA}, 4848.14 \text{ \AA}, 4606.89 \text{ \AA}, 4309.67 \text{ \AA} \text{ \& } 4048.48 \text{ \AA} \text{ respectively.}$$

$$(3) \Delta = (\mu_e \mu_o) t = 1.2 \times 10^{-7} m$$

$$(i) \text{LCP } \delta = \frac{3\pi}{2} \Rightarrow \frac{3\pi}{2} = \frac{2\pi}{\lambda} \Delta \Rightarrow \lambda = 1600 \text{ \AA}$$

$$(ii) \text{RCP } \delta = \frac{\pi}{2} \Rightarrow \lambda = 4\Delta = 4800 \text{ \AA}$$



(a) on rotating N_2 if intensity variation with $I_{\min} = 0$ SGP

(b) if intensity variation is b/w $I_{\max}$ to $I_{\min} \neq 0$, but ^{at different} the position with α to initial. HWP

(c) Either intensity remains same or varies b/w $I_{\max}$ to $I_{\min} \neq 0$ QWP

$$(5) \theta = \frac{\delta}{2} \text{ Here } \delta = \frac{2\pi}{\lambda} (\mu_o \mu_e) t; \text{ given } \delta = \pi/2, \text{ Therefore}$$

$$\frac{2\pi}{\lambda} (\mu_o \mu_e) t = \frac{\pi}{2} \Rightarrow t = 32.37 \mu m$$

$$(6) \theta = -\frac{42^\circ}{20} \times 10 + \frac{27^\circ}{30} \times 20 = -21^\circ + 18^\circ = -3^\circ \text{ ie } 3^\circ \text{ Right handed}$$

(7) From question it is clear that given plate is HWP, so

$$(\mu_o \mu_e) t = (2m+1) \frac{\lambda}{2}; \text{ For least thickness } m=0 \text{ so}$$

$$t = \frac{\lambda}{2(\mu_o \mu_e)} = 32.37 \mu m$$

$$(8) S = \frac{W}{V} \text{ or } C = \frac{9.9}{2 \times 66} = 0.075 \text{ gm/cc ie } 75 \text{ gm/liter } (\because 1 \text{ liter} = 10^3 \text{ cc})$$

It means 75 gm Sugar is pure.

$$\% \text{ of pure sugar} = \frac{75}{80} \times 100 = 93.75\%$$

Tutorial Sheet-6 (Physics-1, 15B11PH111) (Odd Semester, 2024)

Assignment 6: Special Theory of Relativity (STR)

[CO1]

- 1.a)** What are the differences between the inertial and non-inertial frames of reference?
- b)** In which limit Lorentz transformation will be reduced to Galilean transformation?
- c)** What is the difference between the rest mass and the relativistic mass of a particle?
- d)** The rest mass of electron is $0.51 \text{ MeV}/c^2$. Convert it into Kgs.
[Ans.: $9.1 \times 10^{-31} \text{ Kgs.}$]

[CO2]

- 2.a)** The average lifetime of a π meson in its own frame of reference is 26.0 ns. If the π meson moves with speed $0.95c$ with respect to the Earth, (i) what is its lifetime as measured by an observer at rest on Earth? (ii) What is the average distance it travels before decaying as measured by an observer at rest on Earth?
[Ans.: 83.26 ns, 23.72 m]
- b)** Suppose a spaceship is approaching the Earth at half the speed of light and shoots a canister at a speed of $0.75c$. At what velocity will an Earth-bound observer see the canister (i) if it is shot directly towards the Earth? (ii) if it is shot directly away from the Earth?
[Ans.: $0.909 c$, $-0.4 c$]

[CO3]

- 3.a)** An electron's speed is increased from $0.6c$ to $0.8c$. By what ratio does its momentum and kinetic energy change? [Ans.: 1.78, 2.68]
- b)** A proton has a kinetic energy of 950 MeV. Find its momentum according to relativistic mechanics. The mass of proton is $938 \text{ MeV}/c^2$.
[Ans.: $1638.5 \text{ MeV}/c$]

c) Electrons in projection television sets are accelerated through a potential difference of 51 kV. (i) Calculate the speed of the electrons using the relativistic form of kinetic energy assuming the electrons start from rest. Calculate the speed of the electrons using the classical form of kinetic energy. (ii) Is the difference in speed significant in the design of this set? Given the rest mass of electron is $0.51 \text{ MeV}/c^2$.

[Ans.: 0.416 c, 0.447 c]

[CO4]

4. A rod of length L' moves with speed v along the horizontal x' direction in S' frame. The rod makes an angle θ' with respect to the x' axis. (i) Determine the length of the rod as measured by a stationary observer. (ii) Determine the angle θ the rod makes with the x axis.

[Ans.: $L = L'(1 - \beta^2 \cos^2 \theta')^{1/2}$, $\theta = \frac{\tan \theta'}{\sqrt{1 - \beta^2}}$]

Problems pertaining to self-study

1. A box on a shop's rack is measured 70 cm (x axis) $\times$ 35 cm (y axis) $\times$ 25 cm (z axis) by the shop's owner. A traveller takes the suitcase in a train moving with velocity $0.8c$ in the x direction. If the suitcase is placed with its length along the train's velocity, what will be the dimensions measured by the ground observer? **[Ans.: $42 \text{ cm} \times 35 \text{ cm} \times 25 \text{ cm}$]**
2. Find the momentum of an electron (in MeV/c) whose speed is $0.6c$.
3. Evaluate $[1 + p^2/(m^2 c^2)]$, where p is the relativistic momentum and m is the rest mass. **[Ans.: γ^2]**

Q1) (a)Inertial frame

- ① Frame of reference where Newton's first law of inertia holds

Non Inertial frame

- ① The frame of reference where the law of inertia does not hold.

- ② All the non-accelerated frames of reference which move with a constant velocity w.r.t an inertial frame of reference are also inertial frame.

- ② The frames which ~~are~~ are accelerated w.r.t the inertial frame are called non inertial frame.

(b) limit $\boxed{v \ll c} \Rightarrow \gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} = 1$

Lorentz factor

(c) Rest Mass $\rightarrow$ Mass which remains constant regardless of particle's speed or the observer's frame of reference
(m_0)

Relativistic mass $\rightarrow$ Mass which increases with particle speed and is ~~indep~~ depend on observer's frame of reference.
(m)

$\boxed{m = \gamma m_0}$ relation

↙ Mega-electron volts {Unit of Energy}

d) $\frac{\text{MeV}}{c^2} \xrightarrow{\text{to}} \text{Kg} \leftarrow \text{SI unit of mass}$

↳ speed of light
Ka square $(3 \times 10^8 \frac{\text{m}}{\text{sec}})$

Masse Ke Liye :- {Dimensional Analysis} [MLT]
 $\therefore E = mc^2 \Rightarrow m = \frac{E}{c^2}$ (Energy)
 c^2 (speed of light) Ka square

or $1 \text{ MeV} = 1.602 \times 10^{-13} \text{ J}$ $\therefore 1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$
 Mega $\rightarrow 10^6$

$m = \frac{E \text{ (in J)}}{c^2} = \text{To obtain Mass in Kg}$

$E = 0.51 \text{ MeV} = 0.51 \times 1.602 \times 10^{-13} \text{ J}$

$c^2 = (3 \times 10^8 \text{ m/s})^2 = 9 \times 10^{16} \text{ m}^2/\text{s}^2$

$m = \frac{0.817 \times 10^{-13} \text{ J}}{9 \times 10^{16} \text{ m}^2/\text{s}^2}$

$1 \text{ J} = 1 \frac{\text{Kg m}^2}{\text{s}^2}$

$m = \frac{0.817 \times 10^{-13}}{9 \times 10^{16}} \frac{\text{Kg m}^2/\text{s}^2}{\text{m}^2/\text{s}^2}$

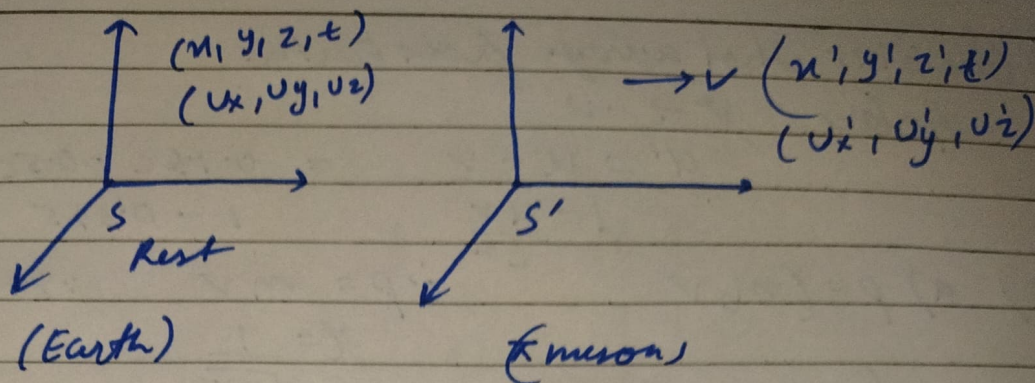
$m = 0.910 \times 10^{-30} \text{ Kg}$

$m = 9.1 \times 10^{-31} \text{ Kg} \rightarrow$

Ratta !!

$1 \text{ MeV} = 1.782 \times 10^{-30} \text{ Kg}$

Ques 2) (a)



(i) $\Delta t = \gamma \Delta t'$

Given $\Delta t' = 26 \text{ ns}$

$v = 0.95c$

$$\Delta t = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} \cdot 26 = \frac{26}{\sqrt{1 - (0.95)^2}} = 83.26 \text{ ns}$$

(ii) Average distance travelled $\Rightarrow v \Delta t'$

$$d = 0.95c \times 83.26 \times 10^{-9} \\ = 23.72 \text{ m}$$

(b) (i) canister shot toward Earth

$u' = \frac{u + v}{1 + \frac{uv}{c^2}}$ Velocity Addⁿ

Given $u = 0.75c$ { canister speed }

$v = 0.5c$ { spaceship speed }

$$u' = \frac{0.75c + 0.5c}{1 + \frac{0.75 \times 0.5 c^2}{c^2}} = \frac{1.25c}{1.375} = 2.72 \times 10^8 \text{ m/sec}$$

(ii) Gunster shot away from Earth

Relace $v \rightarrow -v$

$$u' = \frac{u - v}{1 - \frac{uv}{c^2}} = \frac{0.76c - 0.5c}{1 - 0.375} = 0.4c$$

Que 3) a) $p = \gamma m_0 v$

$$\therefore p = m v$$

$$\therefore m = \gamma m_0$$

$\uparrow$ Rest mass

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$KE = (\gamma - 1) m_0 c^2$$

$$v = 0.6c \Rightarrow \gamma = \frac{1}{\sqrt{1 - (0.6)^2}} = 1.25 \Rightarrow p_1 = 1.25 \times m_0 \times 0.6c$$

$$v = 0.8c \Rightarrow \gamma = \frac{1}{\sqrt{1 - (0.8)^2}} = 1.66 \Rightarrow p_2 = 1.66 \times m_0 \times 0.8c$$

$$\text{Ratio} \Rightarrow \frac{p_2}{p_1} = \frac{1.66 \times 0.8}{1.25 \times 0.6} = 1.776 \approx 1.78$$

$$KE_1 = (1.25 - 1) m_0 c^2 = 0.25 m_0 c^2$$

$$KE_2 = (1.66 - 1) m_0 c^2 = 0.66 m_0 c^2$$

$$\text{Ratio} \Rightarrow \frac{KE_2}{KE_1} = 2.64 \approx 2.68$$

(b) Given $KE = 950 \text{ MeV}$
 $m_0 = \frac{938 \text{ MeV}}{c^2}$

$$\left. \begin{array}{l} T.E \\ \text{(Total Energy)} \\ (E) \end{array} \right\} = KE + m_0 c^2$$

$\downarrow$ Kinetic Energy $\leftarrow$ Rest Mass Energy

$$E = 950 + 938 = 1888 \text{ MeV}$$

$$E = \sqrt{p^2 c^2 + \underbrace{m_0^2 c^4}_{E_0^2}}$$

$$E^2 - m_0^2 c^4 = p^2 c^2 \Rightarrow (pc)^2 = (1888)^2 - (938)^2$$

$$pc = \sqrt{2682300} \text{ MeV}$$

$$\boxed{p = \frac{1638.5 \text{ MeV}}{c}}$$

Ques 3 (c)

Ques 3 (c)

Given $V = 51 \text{ kV} = 51 \times 10^3 \text{ V}$

$$m_0 = 0.51 \frac{\text{MeV}}{c^2}$$

(i) As we know $KE = eV$

$KE = qV$ (Charge $\times$ Potential diff.)

$e \rightarrow \text{charge} \rightarrow 1.602 \times 10^{-19} \text{ C}$

$$KE = 1.602 \times 10^{-19} \times 51 \times 10^3 = 81.7 \times 10^{-16} \text{ J} = 8.1 \times 10^{-15} \text{ J}$$

~~$E(\text{Total Energy}) = KE + m_0 c^2$~~
 ~~$= KE + (0.51 \text{ MeV}) \times c^2$~~
 ~~$= KE + (0.51 \times 1.602 \times 10^{-13}) \text{ J}$~~
 ~~$= 8.17 \times 10^{-15} + 8.19 \times 10^{-14}$~~
 ~~$= 9.01 \times 10^{-14} \text{ J}$~~

~~$m_0 c^2 = 0.51 \text{ MeV} \times (1.602 \times 10^{-13} \frac{\text{J}}{\text{MeV}})$~~
 ~~$= 8.163 \times 10^{-14} \text{ J}$~~

~~$KE = (\gamma - 1) m_0 c^2$~~

~~$KE = \left(\frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} - 1 \right) 0.51 \times 1.602 \times 10^{-13} \text{ J}$~~

~~$8.1 \times 10^{-15} \text{ MeV}$~~

$1 \text{ keV} = 0.001 \text{ MeV}$

$eV = \left(\frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} - 1 \right) m_0 c^2$

$$\Rightarrow \frac{eV + 1}{m_0 c^2} = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$\Rightarrow \left(\frac{eV + 1}{m_0 c^2} \right)^2 = \frac{1}{1 - \frac{v^2}{c^2}}$$

$$\frac{v^2}{c^2} = 1 - \frac{1}{\left(\frac{eV + 1}{m_0 c^2} \right)^2}$$

$v = c \sqrt{1 - \frac{1}{\left(\frac{eV + 1}{m_0 c^2} \right)^2}}$

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$$\frac{eV}{m_e c^2} = \frac{8.1 \times 10^{-15}}{8.16 \times 10^{-14}} \approx 0.1$$

$$V = c \sqrt{1 - \frac{1}{(0.12 + 1)^2}} \Rightarrow c \sqrt{1 - \frac{1}{1.21}} \Rightarrow c \sqrt{0.1736}$$

$$V \approx c \times 0.416$$

$$V = 0.416c$$

(ii) $K.E = \frac{1}{2} m_e v^2 \Rightarrow eV = \frac{1}{2} m_e v^2 \Rightarrow v = \sqrt{\frac{2eV}{m_e}}$

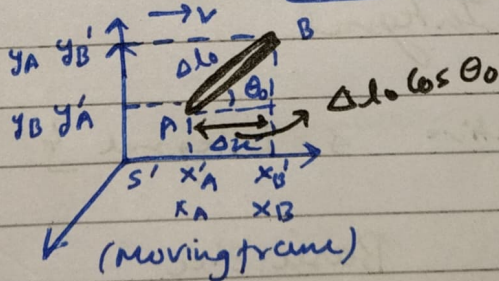
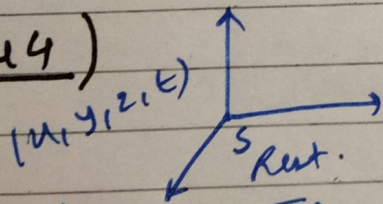
$$v = \sqrt{\frac{2 \times 8.1 \times 10^{-15}}{0.51 \text{ MeV} / c^2}} = \sqrt{\frac{2 \times 8.1 \times 10^{-15}}{9.1 \times 10^{-31} \text{ kg}}}$$

$$0.51 \times 1.602 \times 10^{-13} / (3 \times 10^8)^2$$

$$v = 4.24 \times 10^8 \text{ m/s}$$

$$\Rightarrow V = 1.413c$$

Q4)



$$x_B - x_A = \Delta x$$

$$y_B - y_A = \Delta y$$

$$x'_B - x'_A = \Delta x'$$

$$y'_B - y'_A = \Delta y'$$

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} \Delta x = \Delta x' \sqrt{1 - \frac{v^2}{c^2}} \quad \text{length contraction}$$

$$\Delta x = \Delta x' \gamma \quad \text{Contraction}$$

$$\Delta x = \gamma \Delta x' \quad \text{Contraction}$$

$$\Delta y = \Delta y' \quad \text{As velocity is only along x-axis so No length contraction along y}$$

$$\Delta l^2 = (\Delta x')^2 + (\Delta y')^2 \quad \text{--- (3)}$$

$$\Delta l^2 = \Delta x^2 + \Delta y^2 \quad \text{--- (4)}$$

$$\text{Solving (1), (2), (3) \& (4)} \Rightarrow \Delta l^2 = \Delta x'^2 \left(1 - \frac{v^2}{c^2}\right) + (\Delta y')^2 = (\Delta x')^2 \frac{\Delta x'^2 v^2 + \Delta y'^2}{c^2}$$

$$\Delta x' = \Delta l \cos \theta_0 \Rightarrow \Delta l = \Delta l_0 - \Delta l_0^2 \cos^2 \theta_0 \frac{v^2}{c^2} \Rightarrow \Delta l = \Delta l_0 \sqrt{1 - \frac{v^2 \cos^2 \theta_0}{c^2}}$$

$$\therefore \tan \theta_0 = \frac{\Delta y'}{\Delta x'} \neq \frac{\Delta y}{\Delta x} \quad \text{--- (5)}$$

$$\tan \theta = \frac{\Delta y}{\Delta x} = \frac{\Delta y'}{\Delta x' \sqrt{1 - \frac{v^2}{c^2}}} = \frac{\tan \theta_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Tutorial Sheet – 7
B. Tech. First Semester (2024-25)
Physics – 1 (15B11PH111)

Problems:

1. Write down the expression of the total relativistic energy for the massless and massive particles. [CO1]
2. The total energy of a moving particle is exactly (a) two times (b) equal to its rest energy. Find the speed of the particle in both cases. [CO3]
3. A proton has a relativistic energy as 900 MeV. Comment on the state of motion of the proton. The rest mass of a proton is 1.6×10^{-27} kg. Discuss validity of the above situation. [CO3]
4. Is a black body black? Is a mercury lamp black body? Show that the Rayleigh-Jeans Law and the Wien's Law are consistent with Planck radiation law in appropriate limit. Discuss briefly the ultraviolet catastrophe. [CO2]
5. Two bodies A and B have thermal emissivity of 0.01 and 0.81, respectively. The outer surface area of the two bodies are the same. The two bodies emit total radiant power at the same rate. If the temperature of A is 5802 K, calculate a) temperature of B, b) maximum emitted wavelength of A, and c) maximum emitted wavelength of B. [CO3]
6. An ideal blackbody at temperature T, emits radiation of energy density u. The corresponding value for a material at temperature T/2 is u/256. Calculate the emissivity of the material. [CO3]
7. The peak wavelength, at which maximum intensity is observed in the radiation spectrum of sun is 483 nm. Estimate the temperature of the surface of sun. [CO3]
8. A body emits 15 W/m^2 energy at temperature 27°C . What should be the temperature of the same object so that it emits 240 W/m^2 energy? [CO3]

Practice questions:

1. A body have surface area of 5 sq. cm and temperature of 727°C . If the body emits 300 J energy per minute determine emissivity of the body. [CO3]
2. The earth receives 1.4 kW of power from the sun. Assume that both earth and the sun to be black bodies. If the radius of the sun is $6.95 \times 10^8 \text{ m}$ and the earth-sun distance is $1.5 \times 10^{11} \text{ m}$, calculate the temperature of the sun. [CO3]
3. The power radiated by a blackbody is P and it radiates maximum energy at wavelength, λ_0 . If the temperature of the blackbody is now changed so that it radiates maximum energy at wavelength $\frac{3}{4}\lambda_0$, the power radiated by it becomes nP. Find the value of n. [CO3]

PHY:- Tute-7 2401020065 "Utkarsh Gupta."

Ques 1)

$$\text{Total Relativistic Energy} \Rightarrow E = \sqrt{m_0^2 c^4 + p^2 c^2}$$

$$\text{For massless particle } m_0 = 0 \Rightarrow E = \sqrt{p^2 c^2} = pc$$

Ques 2) (a) $E = 2E_0$

$$2E_0 = \sqrt{E_0^2 + p^2 c^2}$$

$$4E_0^2 = E_0^2 + p^2 c^2$$

$$\Rightarrow 3E_0^2 = p^2 c^2$$

$$\Rightarrow \frac{\sqrt{3} E_0}{c} = p$$

$$\text{Now } p = \gamma m_0 v \quad \text{and } E_0 = m_0 c^2$$

$$\Rightarrow \frac{\sqrt{3} E_0}{m_0 v c} = \gamma \quad \Rightarrow \frac{\sqrt{3} m_0 c^2}{m_0 v c} = \gamma$$

$$\Rightarrow \frac{\sqrt{3} c}{v} = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$\Rightarrow \sqrt{1 - \frac{v^2}{c^2}} = \frac{v}{\sqrt{3} c} \Rightarrow 1 - \frac{v^2}{c^2} = \frac{v^2}{3c^2} \Rightarrow 1 = \frac{v^2}{c^2} \left[1 + \frac{1}{3} \right]$$

$$1 = \frac{4}{3} \frac{v^2}{c^2}$$

$$\Rightarrow \frac{3 \times c^2}{4} = v^2 \Rightarrow \boxed{\frac{\sqrt{3} c}{2} = v}$$

(b) $E = E_0$; $E_0^2 = E_0^2 + p^2 c^2$

$$\boxed{\begin{matrix} p=0 \\ v=0 \end{matrix}}$$

Que 3)

$$E = 900 \text{ MeV}$$

$$m_0 = 1.6 \times 10^{-27} \text{ kg}$$

$$E = m_0 c^2$$

$$= 1.6 \times 10^{-27} \times \underbrace{3 \times 10^8 \times 3 \times 10^8}_{9 \times 10^{16}}$$

$$= 1.44 \times 10^{-10} \text{ J}$$

In MeV $\Rightarrow$

$$1 \text{ MeV} = 1.6 \times 10^{-19} \text{ J}$$

$$1 \text{ J} = \frac{1}{1.6 \times 10^{-19}} \text{ MeV}$$

$$1.44 \times 10^{-10} \text{ J} \longrightarrow \frac{1.44 \times 10^{-10}}{1.6 \times 10^{-19}} \text{ MeV}$$

$$E_0 = 900 \text{ MeV}$$

As $E = E_0 \rightarrow$ particle is at rest i.e. stationary

Que 4)

$\rightarrow$ black body absorbs all incident radiation but not necessarily black

$\rightarrow$ No Mercury lamp is not Black Body
 $\hookrightarrow$ It emits radiations at discrete wavelengths unlike Black Body which emits constant spectrum.

$\rightarrow$ Refer Lec 31-32-33

Ques 5)

$$e_A = 0.01 \quad e_B = 0.81$$

$$T_A = 5802 \text{ K} \quad T_B = ?$$

Stefans Boltzmann Law { Lec 32 }

$$E = \sigma T^4$$

(a) temp of B

$$e_A T_A^4 = e_B T_B^4 \quad \text{[Acc to Ques]}$$

$$\left(\frac{0.01 \times (5802)^4}{0.81} \right)^{\frac{1}{4}} = T_B \Rightarrow \boxed{T_B = 1934 \text{ K}}$$

(b) $(\lambda_A)_{\max}$

Weins displacement law $\lambda_m T = \text{constant} = 2.898 \times 10^{-3} \text{ m.K}$ (b)

$$\lambda_{\text{peak}} T = 2.898 \times 10^{-3} \text{ m.K}$$

$$(\lambda_A)_{\max} = \frac{2.898 \times 10^{-3} \text{ m.K}}{5802 \text{ K}} = 500 \text{ nm}$$

(c) $(\lambda_B)_{\max}$

$$(\lambda_B)_{\max} = \frac{2.898 \times 10^{-3}}{1934} = 1500 \text{ nm}$$

Que 6)

Stefans Law

$$E = e T^4$$

$\therefore e = 1$ For ^{ideal} Black Body

$$T_1 = T \quad ; \quad E_1 = 4 \quad ; \quad e_1 = 1$$

$$T_2 = \frac{T}{2} \quad ; \quad E_2 = \frac{4}{256} \quad ; \quad e_2 = e_1$$

$$\frac{e_2 (T_2)^4}{e_1 (T_1)^4} = \frac{E_2}{E_1} = \frac{4}{4 \times 256} = \frac{(1) \times (T/2)^4}{(1) T^4}$$

$$\frac{1}{256} = e_2 \frac{T^4}{16 T^4} \Rightarrow \frac{16}{256} = e_2$$

$$e_2 = 0.0625$$

Que 7)

$$\lambda_{\text{peak}} \times T = 2.9 \times 10^{-3} \text{ m} \cdot \text{K}$$

$$T = \frac{2.9 \times 10^{-3}}{4.83 \times 10^{-9}}$$

$$\Rightarrow 6000 \text{ K}$$

Que 8)

$$E_1 = 15 \frac{\text{W}}{\text{m}^2}$$

$$E_2 = 240 \frac{\text{W}}{\text{m}^2}$$

$$T_1 = 27^\circ \text{C}$$

$$T_2 = ?$$

$$\text{in Kelvin} = 273 + 27 = 300 \text{ K}$$

$$E \propto T^4$$

$$\frac{E_1}{E_2} = \left(\frac{T_1}{T_2} \right)^4$$

$$\left(\frac{15}{240} \right)^{1/4} \times \frac{1}{300 \text{ K}} = \frac{1}{T_2} \Rightarrow T_2 = 600 \text{ K}$$

TUTORIAL SHEET 8, PHYSICS-1 (ODD-2024)

1. What is the difference between a mechanical wave and de Broglie wave? [CO1]
2. What are the phase velocity and the group velocity? [CO1]
3. Explain why we don't observe wave characteristics of a golf ball (typical mass of a golf ball is 46 grams)? [CO2]
4. Calculate the de Broglie wavelength of a cricket ball ($m = 156$ gm), and an electron ($m = 9.1 \times 10^{-31}$ kg) both moving at 100 m/s? [CO4]
5. Show that the group velocity is given by $v_g = -\hbar \frac{dv}{d\lambda}$. [CO3]
6. The phase velocity of ripples on a liquid surface is $\sqrt{\frac{2\pi S}{\lambda \rho}}$, where S is the surface tension and ρ is the density of the liquid. Find the group velocity of the ripples. [CO3]
7. Calculate the de Broglie wavelength of a neutron ($m_N = 1.6 \times 10^{-27}$ kg) moving with speed 0.8c. [CO4]
8. Find the kinetic energy and the phase velocity of de Broglie wave in the problem number 7. What would be the group velocity in this case? [CO4]

Self study problems:

1. The phase velocity of the waves on the sea surface is $\sqrt{\frac{g\lambda}{2\pi}}$, where g is acceleration due to gravity. Find the group velocity of the waves. [CO3]
2. Find the de Broglie wavelength of neutron having kinetic energy equal to 1.00 MeV. Is the relativistic calculation needed? [CO4]
3. Can a phase velocity be equal to the group velocity? Give examples. [CO2]

TUTORIAL: 8 (SOLUTIONS)

1. Mechanical wave is a disturbance in an elastic medium. de-Broglie waves are associated with the moving particles. In the conventional interpretation these waves are the measure of position probabilities.

2. Refer notes / Lecture slides.

3. de Broglie wavelength is very small compared to the physical size.

4.

$$\lambda_{co} = \frac{6.63 \times 10^{-34}}{1.56 \times 10^2 \times 100} \approx 4.25 \times 10^{-36} \text{ m}$$

$$\lambda_c = \frac{6.63 \times 10^{-34}}{9.1 \times 10^{-31} \times 100} \approx 0.7285 \times 10^{-5} \text{ m}$$

5.

$$v_g = \frac{d\omega}{dk}, \quad \omega = 2\pi\nu \quad \text{and} \quad k = \frac{2\pi}{\lambda}$$

$\downarrow d\omega = 2\pi d\nu$ $\Rightarrow dk = -\frac{2\pi}{\lambda^2} d\lambda$

$$v_g = -\lambda^2 \frac{d\nu}{d\lambda}$$

6.

$$v_p = \sqrt{\frac{2\pi S}{\lambda g}} = \sqrt{\frac{k S}{g}}$$

$$v_p = \frac{\omega}{k} = \sqrt{\frac{k S}{g}}$$

$$d\omega = \frac{3}{2} \sqrt{\frac{k S}{g}} dk$$

$$v_g = \frac{d\omega}{dk} = \frac{3}{2} \sqrt{\frac{k_B}{g}}$$

$$\boxed{v_g = \frac{3}{2} v_F}$$

$$7. \quad \lambda = \frac{h}{\gamma m_N v}, \quad \gamma = \frac{1}{\sqrt{1 - v^2/c^2}}$$

$$\lambda = \frac{6.63 \times 10^{-34}}{\sqrt{1 - 0.8^2} \times 1.5 \times 10^{-27} \times 0.8 \times 3 \times 10^8}$$

$$\lambda = 2.8776 \times 10^{-15} \text{ m}$$

8.

$$p_c = \frac{hc}{\lambda}$$

$$= \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{2.8776 \times 10^{-15}}$$

$$p_c = 6.912 \times 10^{-11} \text{ kg } \frac{\text{m}}{\text{s}^2}$$

(i) Kinetic Energy, $T = E - m_0 c^2 = E - E_0$

$$= \sqrt{(pc)^2 + (m_0 c^2)^2} - m_0 c^2$$

$$E_0 = 9.1 \times 10^{-31} \times (3 \times 10^8)^2$$

$$E_0 = 8.1 \times 10^{-15} \text{ J}$$

$$E_0 = 1.6 \times 10^{-27} \times (3 \times 10^8)^2$$

$$E_0 = 1.44 \times 10^{-10} \text{ J}$$

$$T = \sqrt{(6.912 \times 10^{-11})^2 + (1.44 \times 10^{-10})^2}$$

$$T = 1.57297 \times 10^{-11} \text{ J}$$

(ii) Phase velocity:

Let v be the particle velocity.

$$E = \frac{E_0}{\sqrt{1 - v^2/c^2}}$$

$$E = 1.57297 \times 10^{-11} + 1.44 \times 10^{-10}$$

$$E = 1.5572 \times 10^{-10} \text{ J}$$

$$v = c \sqrt{1 - \frac{E_0^2}{E^2}}$$

$$= 3 \times 10^8 \sqrt{1 - \frac{1.44 \times 10^{-10}}{1.5572 \times 10^{-10}}}$$

$$\gamma = 0.823c$$

$$\gamma_p = \frac{c}{v} = \frac{0.823c}{c} = 1.215c$$

$$\gamma_g = \gamma = 0.823c$$

Self study problems:

Page 4

$$\textcircled{1} \quad v_p = \frac{\omega}{k} = \sqrt{\frac{g\lambda}{2\pi}} = \sqrt{\frac{g}{k}}$$

$$d\omega = \frac{1}{2} \sqrt{\frac{g}{k}} dk$$

$$v_g = \frac{d\omega}{dk} = \frac{1}{2} \sqrt{\frac{g}{k}} = \frac{1}{2} v_p$$

$$\textcircled{2} \quad E_k = 1 \text{ MeV} = 1.6 \times 10^{-13} \text{ J} = \text{kinetic energy}$$

Since, Rest energy $= m_N c^2 = 937.5 \text{ MeV}$,

$E_k \ll E_{\text{rest}}$ & we can use non-relativistic expressions,

$$E_k = \frac{p^2}{2m_N}$$

$$p = \sqrt{2E_k m_N}$$

$$= \sqrt{2 \times 1.6 \times 10^{-13} \times 1.67 \times 10^{-27}}$$

$$p = 2.311 \times 10^{-20} \text{ kg m/s}$$

$$\lambda = \frac{h}{p} = \frac{6.63 \times 10^{-34} \text{ J}\cdot\text{s}}{2.311 \times 10^{-20} \text{ kg m/s}}$$

$$\lambda = 28.6 \times 10^{-15} \text{ m}$$

Tutorial 9 : Physics 1 (15B11PH111), ODD 2024

Assignment 4: Nonexistence of electrons in the atomic Nucleus

Assignment 5: Particle in a three-dimensional box

[CO1]

1. a) A particle is trapped in a box of size L . What is the uncertainty in its position Δx ?
b) Why does tunnelling occur, even if a particle's energy is less than the potential barrier?
2. a) Show the difference in the particle with lower localization and higher localization by drawing the wave packets.
b) What is the limit for the perfect localization or the limit for classical particles?
3. If the uncertainty in the measurement of position is infinity for a particle, then what will be the uncertainty in the measurement of momentum?

[CO2]

4. The electron position in an atom has an uncertainty $\pm r$. The uncertainty in its velocity therefore is $\pm v$. Taking $\Delta x = r \approx 0.5 \times 10^{-10} \text{ m}$, find the uncertainty in the electron velocity. (Ans: $1.15 \times 10^6 \text{ m/s}$)
5. Uncertainty in time of an excited atom is about 10^{-8} s . What are the uncertainties in energy and in frequency of the radiation emitted by an atom? (Ans: $7.96 \times 10^6 \text{ Hz}$)
6. Draw the wavefunction and probability distribution for a particle in a box at the $n=1, 2$ and 3 energy levels. Identify the regions for each n where particle is never found or less likely to be found.

[CO3]

7. An electron of energy 350 eV is passed through a circular hole of radius 10^{-7} m . What is the uncertainty introduced in the angle of emergence? (Ans: $5.22 \times 10^{-5} \text{ radians}$)
8. The de-Broglie wavelength associated with the electron moving back and forth in the infinite potential well in the first excited state is $3 \times 10^{-8} \text{ m}$. Calculate the de-Broglie wavelength associated with the electron in ground, second, and third excited state. (Ans: $\lambda_1 = 6 \times 10^{-8} \text{ m}, \lambda_3 = 2 \times 10^{-8} \text{ m}, \lambda_4 = 1.5 \times 10^{-8} \text{ m}$)

[CO4]

9. A stream of electrons, each of energy $E = 3 \text{ eV}$, is incident on a potential barrier of height $V = 4 \text{ eV}$. The width of the barrier is $20 \text{ \AA}$. Calculate the percentage transmission of the beam through this barrier. (Ans: 10^{-28})
10. A particle limited to the x-axis has the wave function $\psi = ax$ between $x = 0$ and $x = 1$; $\psi = 0$ elsewhere. Find the probability that the particle can be found between $x = 0.25$ and $x = 0.50$. (Ans: 10.9%)

Extra practice questions:

1. An electron and a 150 gm baseball are traveling at a velocity of 220 m/s , measured to an accuracy of 0.065% . Calculate and compare uncertainty in position of each. (Ans: $0.4 \text{ mm}, 2.5 \times 10^{-33} \text{ m}$)
2. Explain the difference between time dependent and time independent Schrodinger equation.
3. An α particle with energy $E = 6 \text{ MeV}$, is trapped in a nucleus of radius $R = 5 \text{ fm}$. The potential barrier at the surface of the nucleus is $V = 12 \text{ MeV}$. (Ans: 2.24×10^{-5})
4. An electron is confined to move in a one-dimensional potential well of length $5 \text{ \AA}$. Find the quantized energy values for the three lowest energy states. (Ans: $E_1 = 15.5 \text{ eV}, E_2 = 62 \text{ eV}, E_3 = 139.5 \text{ eV}$)

Solutions, Tutorial 9, Physics 1 (15B11PH111) - 0002024

[1(a)] $\Delta x = L$

[1(b)] wavelike nature of the particles, allowing finite probability of transmission.

[2(a)] Refer notes.

[2(b)] $\hbar \rightarrow 0$

(3) $\Delta x \rightarrow \infty$, $\Delta x \Delta p \geq \hbar/2$
 $\Rightarrow \Delta p \sim \frac{\hbar}{2\Delta x} \Rightarrow \underline{\Delta p = 0}$

(4) $\Delta x \Delta p \geq \hbar/2$

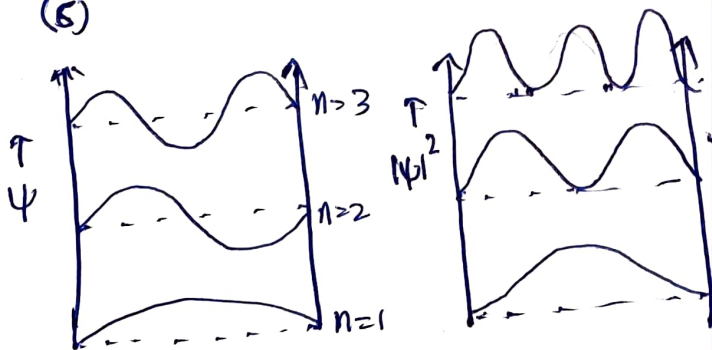
$\Delta p \sim m \Delta v$, $m \Delta x \Delta v \geq \hbar/2$

$\Delta v \approx \frac{\hbar}{2m \Delta x} = \frac{h}{4\pi m \Delta x}$

$\Delta v = \frac{6.62 \times 10^{-34} \text{ Js}}{4 \times 3.14 \times 9.11 \times 10^{-31} \text{ kg} \times 0.5 \times 10^{-10} \text{ m}}$

$\underline{\Delta v = 1.15 \times 10^6 \text{ m/s}}$

(5)



$n=0 \Rightarrow$ particle is less likely to be found at the end of the wall.

$n=1, 2 \Rightarrow$ particle is never found at nodes.

For $n=1$: there is one node at the center

for $n=2$: 2 nodes,

(5) $\Delta E \Delta t \geq \hbar/2$, $\Delta E = h \Delta f$
 $\Delta t = 10^{-8} \text{ s}$, $\hbar = \frac{h}{2\pi} = 1.055 \times 10^{-34} \text{ Js}$

$\Delta E \approx \frac{\hbar}{2\Delta t} \approx 5.275 \times 10^{-27} \text{ J}$

$\Delta f = \frac{\Delta E}{h} = \frac{5.275 \times 10^{-27} \text{ J}}{6.62 \times 10^{-34} \text{ Js}}$

$\underline{\Delta f = 7.96 \times 10^6 \text{ Hz}}$

(7) $p = \sqrt{2mE}$

$= \sqrt{2 \times 9.11 \times 10^{-31} \times 350 \times 1.6 \times 10^{-19}}$

$p = 1.01 \times 10^{-23} \text{ kg m/s}$

$\Delta x \Delta p_y \geq \hbar/2$, Δx : slit width,

$\Delta p_y \approx \frac{\hbar}{2\Delta x} \approx \frac{1.055 \times 10^{-34} \text{ Js}}{2 \times 10^{-7} \text{ m}}$

$\Delta p_y \approx 5.275 \times 10^{-28} \text{ kg m/s}$

Uncertainty in the angle $\Delta \theta$

$\Delta \theta \approx \frac{\Delta p_y}{p} \approx \frac{5.275 \times 10^{-28}}{1.01 \times 10^{-23}}$

$\underline{\Delta \theta \approx 5.22 \times 10^{-5} \text{ radian}}$

(8) de Broglie wavelength in the n th energy state is,

$\lambda_n = \frac{2L}{n}$

$\lambda_2 = 3 \times 10^{-8} \text{ m}$, $\lambda_2 = \frac{2L}{2} = L$

$\Rightarrow L = 3 \times 10^{-8} \text{ m}$

Ground state, ($n=1$): $\lambda_1 = \frac{2L}{1} = 2L = 6 \times 10^{-8} \text{ m}$

Second excited state ($n=3$), $\lambda_3 = \frac{2L}{3} = 2 \times 10^{-8} \text{ m}$

3rd excited state ($n=4$), $\lambda_4 = \frac{2L}{4} = 1.5 \times 10^{-8} \text{ m}$

$$(9) T \approx e^{-2KL}, \quad L = 20 \times 10^{-10} \text{ m}$$

$$K = \sqrt{\frac{2m(V-E)}{\hbar^2}}, \quad V = 4 \text{ eV}, E = 3 \text{ eV}$$

$$= \sqrt{\frac{2 \times 9.11 \times 10^{-31} \times 1.0 \times 10^{-19}}{1.113 \times 10^{-68}}}$$

$$K = 1.62 \times 10^{10} \text{ m}^{-1}$$

$$T \approx e^{-2KL} \approx e^{-2 \times 1.62 \times 10^{10} \times 20 \times 10^{-10}}$$

$$T \approx e^{-64.8} \approx 1.0 \times 10^{-28}$$

$$(10) \psi(x) = ax \quad \text{for } 0 \leq x \leq 1$$

$$= 0 \quad \text{elsewhere.}$$

$$\int |\psi|^2 dx = 1, \quad \int_0^1 a^2 x^2 dx = 1$$

$$a^2 \left[\frac{x^3}{3} \right]_0^1 = 1 \Rightarrow a = \sqrt{3}$$

$$\text{Normalised wavefunction, } \psi = \sqrt{3}x$$

$$\text{Probability, } P = \int_{0.25}^{0.5} |\psi|^2 dx$$

$$P = 3 \int_{0.25}^{0.5} x^2 dx = 3 \left[\frac{x^3}{3} \right]_{0.25}^{0.5}$$

$$= [0.5^3 - 0.25^3]$$

$$P = 0.109$$

$$\text{Probability} \approx 10.9\%$$

~~Extra~~ Extra practice problems:

(1) the uncertainty in velocity,

$$\Delta v = v \times (0.065\%)$$

$$= 220 \text{ m/s} \times 0.065/100$$

$$\Delta v = 0.143 \text{ m/s}$$

For electron,

$$\Delta x_e = \frac{\hbar}{2m\Delta v} = \frac{1.05 \times 10^{-34} \text{ Js}}{2 \times 9.11 \times 10^{-31} \times 0.143}$$

$$= 0.4 \text{ nm}$$

For baseball,

$$\Delta x_b = \frac{\hbar}{2m\Delta v} = 2.5 \times 10^{-33} \text{ m}$$

$$\Delta x_e \gg \Delta x_b$$

$$(2) i\hbar \frac{\partial \psi(x,t)}{\partial t} = \hat{H} \psi(x,t) - \text{TISE}$$

$\psi(x,t)$ - wavefunction depends on time

$$\hat{H} \psi(x) = E \psi(x)$$

$\psi(x)$: wavefunction is independent of time.

$$(3) P \approx e^{-2KR}, \quad R = 5 \text{ fm}$$

$$= 5 \times 10^{-15} \text{ m}$$

$$K = \sqrt{\frac{2m(V-E)}{\hbar^2}}$$

$$KR = 5.355, \quad P \sim e^{-2KR}$$

$$P = 2.24 \times 10^{-5}$$

$$(4) E_n = \frac{n^2 h^2}{8mL^2}, \quad L = 5 \times 10^{-10} \text{ m}$$

$$m = 9.109 \times 10^{-31} \text{ kg}$$

$$h = 6.62 \times 10^{-34} \text{ Js}$$

$$E_n = n^2 \cdot 15.5 \text{ eV}$$

$$\text{For } n=1 \Rightarrow E_1 = 15.5 \text{ eV}$$

$$\text{for } n=2 \Rightarrow E_2 = 4 \times 15.5 \text{ eV} = 62 \text{ eV}$$

$$\text{for } n=3 \Rightarrow E_3 = 9 \times 15.5 \text{ eV} = 139.5$$

B. Tech. First Semester (2024-25)

Tutorial Sheet – 10

Course: Physics – I (15B11PH111)

1. A beam of electrons is incident on a barrier 6 eV high and 0.2 nm wide. Find the energy they should have, if 1% of them are to get through the barrier.
2. For a given set of data [electron energy = 1 eV, potential barrier height = 10 eV, width of barrier = 0.5 nm], in which case the tunneling probability of electrons is affected most?
 - (i) Energy of electrons is doubled or
 - (ii) Width of potential barrier is doubled
3. How is it impossible for $2P_{5/2}$ and $2P_{7/2}$ to exist?
4. Find the percentage difference between L and the maximum value of L_z for an atomic electron in p, d, and f states
5. If atoms could contain electrons with principal quantum numbers up to and including $n = 6$ only, how many elements would there be in the periodic table?
6. Applying the split orbit coupling, find the total angular momentum of hydrogen atom in $l = 2$ state.
7. An alpha particle is trapped in a nucleus of radius 1.4×10^{-15} m. What is the probability that it will escape from the nucleus if its energy is 2 MeV? The potential barrier at the surface of the nucleus is 4 MeV and the mass of the α -particle = 6.64×10^{-27} kg.

Soln: Tut 10

① Tunneling Prob

$$T = \frac{1}{100} = e^{-2\beta L}$$

$$\Rightarrow \beta = \frac{1}{2L} \ln\left(\frac{1}{T}\right) = 1.15 \times 10^{10}$$

$$\frac{\sqrt{2m(V_0 - E)}}{\hbar} = \beta$$

$$\Rightarrow V_0 - E = 8.12 \times 10^{-19}$$

$$E = 0.925 \text{ eV}$$

② $T = e^{-2\beta L}$

$$\beta = \frac{\sqrt{2m(V_0 - E)}}{\hbar}$$

$$E = 1 \text{ eV}, V_0 = 10 \text{ eV}$$

$$L = 0.5 \text{ nm}$$

$$T_1 = 1.1 \times 10^{-7}$$

① when E is doubled

$$\Rightarrow T_2 = 2.4 \times 10^{-7}$$

② when width of potⁿ barrier is doubled

$$T_3 = 1.3 \times 10^{-14}$$

$\Rightarrow$ as V increases $T \downarrow$

as $E \uparrow T \uparrow$

③ 2^2P state

$$l=1, N=2S+1=2$$

$$\Rightarrow S=1/2$$

$$J = l \pm S \Rightarrow J = \frac{3}{2}, \frac{1}{2}$$

So $5/2, 7/2$ does not exist

③ $L = \sqrt{l(l+1)} \hbar$

$$L_2 = m \hbar$$

$$L_{2\text{max}} = l \hbar$$

(2nd part at rec end)

So % changes

$$\frac{L - L_2}{L} = \frac{\sqrt{l(l+1)} \hbar - l \hbar}{\sqrt{l(l+1)} \hbar}$$

$$= 1 - \frac{l}{\sqrt{l(l+1)}}$$

for $l=1$ % = 29%

for $l=2$ % = 18%

for $l=3$ % = 13%

④ Total elements

$$= 2(n)^2 + 2(n-1)^2 + 2(n-2)^2 + \dots + 2(1)^2$$

$$n=6$$

$$= 2[6^2 + 5^2 + 4^2 + 3^2 + 2^2 + 1^2] = 182$$

⑤

④ for H atoms in $l=2$ state
 $s = \frac{1}{2}$ $J = l \pm s = \frac{3}{2}, \frac{5}{2}$
 $J = \sqrt{j(j+1)} \hbar$
 $\Rightarrow J = \sqrt{15} \hbar/2, \sqrt{35} \hbar/2$

⑤ $T = 16 \left(\frac{E}{V_0} \right) \left(1 - \frac{E}{V_0} \right) \exp \left(-\frac{2}{\hbar} a \sqrt{2m(V_0 - E)} \right)$

$T = 16 \times \frac{1}{2} \times \frac{1}{2} \times \exp(-3.477)$ $\left\{ \begin{array}{l} a = 1.4 \times 10^{-15} \\ E = 2 \text{ MeV} \\ V = 4 \text{ MeV} \\ m = 6.6 \times 10^{-27} \end{array} \right.$

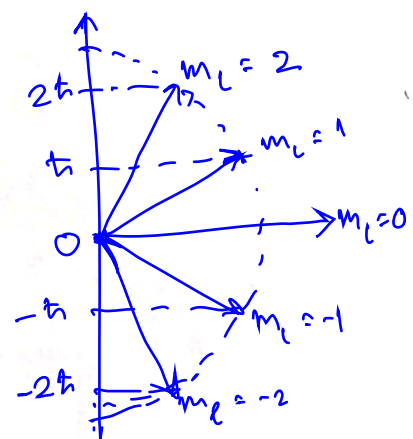
$T = 0.124 \text{ Ans}$

③ second part

$L = \sqrt{l(l+1)} \hbar$
 $L_z = m_l \hbar; m_l = -l, \dots, 0, \dots, l$

$\cos \theta = \frac{L_z}{L} = \frac{m_l}{\sqrt{l(l+1)}}$

$S = \sqrt{s(s+1)} \hbar; s = \frac{1}{2}$
 $S_z = m_s \hbar, m_s = \pm \frac{1}{2}$
 $\cos \theta = \pm \frac{1}{\sqrt{3}}$



For d orbital
 $(l=2)$

s: $l=0 \rightarrow \theta = 0$

p: $l=1 \rightarrow \theta = 0, \cos^{-1}(\pm \frac{1}{\sqrt{2}})$

d: $l=2 \rightarrow \theta = 0, \cos^{-1}(\pm \frac{1}{\sqrt{6}}), \cos^{-1}(\pm \frac{2}{\sqrt{6}})$